

# Yishun Junior College H1 Chemistry Preliminary Examinations 2013

## Paper 1 Answers

|     |      |      |      |      |      |
|-----|------|------|------|------|------|
| 1 A | 6 A  | 11 C | 16 D | 21 C | 26 A |
| 2 B | 7 C  | 12 B | 17 C | 22 B | 27 B |
| 3 D | 8 A  | 13 A | 18 C | 23 D | 28 D |
| 4 D | 9 D  | 14 A | 19 D | 24 A | 29 C |
| 5 C | 10 B | 15 B | 20 A | 25 B | 30 B |

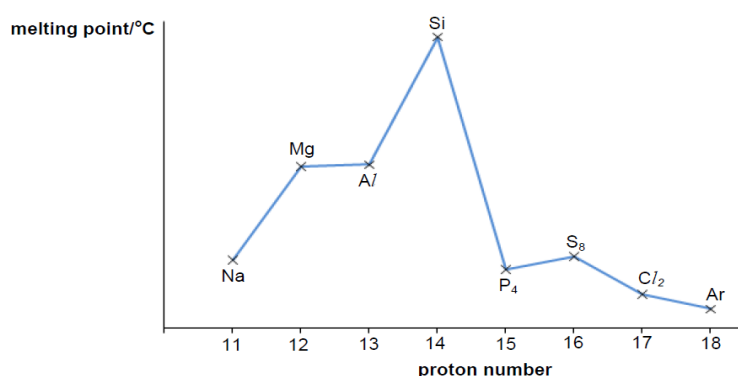
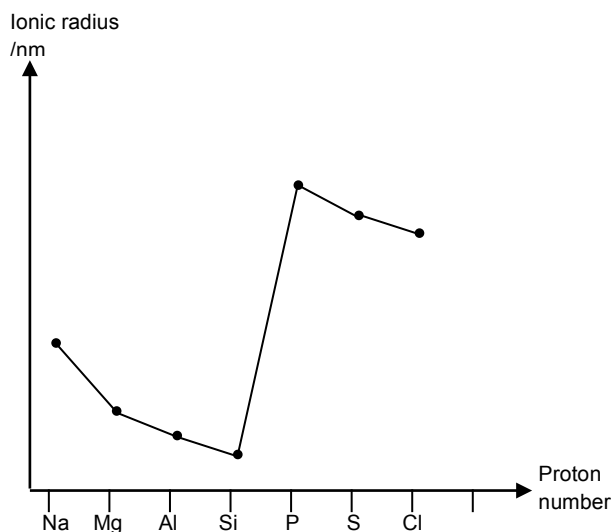
## Paper 2 Answers

- 1 (a) (i) Enthalpy change of combustion of propane is the energy evolved when one mole of propane is completely burnt in oxygen. [1]
- (ii) No. of moles of propane =  $\frac{100}{1000} \div 24 = 4.167 \times 10^{-3}$   
 Heat absorbed by water =  $250 \times 4.18 \times 6 = 6270 \text{ J}$  [1]  
 Heat evolved =  $\frac{6270}{70} \times 100 = 8957 \text{ J}$  [1]  
 $\Delta H_c = -\frac{8957 \times 10^{-3}}{4.267 \times 10^{-3}} = -2150 \text{ kJ mol}^{-1}$  [1] [3]
- (iii)  $\Delta H_f(\text{propane}) = (-394 \times 3) + (-286 \times 4) - (-2150)$  [1]  
 $= -173 \text{ kJ mol}^{-1}$  [1] [2]
- (b) (i)  $1s^2 2s^2 2p^6 3s^2$  [1]
- (ii)  $A_r = \frac{(78.6 \times 24) + (10.11 \times 25) + (11.29 \times 26)}{100} = 24.3$  [1]
- (c)
- |                | Sb    | Cl   |
|----------------|-------|------|
| Mass ratio     | 53.6  | 46.6 |
| Mole ratio     | 0.438 | 1.31 |
| Simplest ratio | 1     | 3    |
- Empirical formula:  $\text{SbCl}_3$  [1]
- $n[(122 + 3(35.5))] = 228.5$   
 $n = 1$   
 Molecular formula:  $\text{SbCl}_3$  [1] [2]
- (d)  $\text{SbCl}_3$  has a simple covalent structure and so, less energy is needed to overcome the weak van der Waals' forces between its molecules. [1]  
 $\text{MgCl}_2$  has a giant ionic lattice structure and so, a lot of energy is needed to overcome the strong ionic bonds. [1] [2]

[Total:12]

- 2 (a) (i)  $C_{21}H_{20}O_3Cl_2$  [1]
- (ii) One of the carbon atoms involved in the  $C=C$  double bond is bonded to 2 similar groups (chlorine atoms) [1]
- (b) Functional group: alkene  
 Type of reaction: addition OR oxidation  
 Reagents and conditions: conc.  $H_2SO_4$ , followed by  $H_2O$  and warm  
 OR  $KMnO_4$ , aq.  $NaOH$  and cold
- Functional group: halogen derivative  
 Type of reaction: substitution  
 Reagents and conditions: aq  $NaOH$ , heat.
- Each functional group and type of reaction [1]  
 Each reagents and conditions [1] [4]
- (c) Replace the chlorine atoms with fluorine atoms as the  $C-F$  bond is very strong and difficult to break. [1] [1]
- (d) (i) Substitution [1]
- (ii) **Step 1**  $X_2(g)$  in  $CCl_4$ , UV light [1]  
**Step 2**  $NaCN$  in ethanol, heat under reflux [1] [2]  
 [Total:10]

- 3 (a) [1] for each sketch



**(b) Ionic radius**

Cations ( $\text{Na}^+$ ,  $\text{Mg}^{2+}$ ,  $\text{Al}^{3+}$ ,  $\text{Si}^{4+}$ ) are isoelectronic –  $1s^2 2s^2 2p^6$ , and anions ( $\text{P}^{3-}$ ,  $\text{S}^{2-}$ ,  $\text{Cl}^-$ ) are isoelectronic –  $1s^2 2s^2 2p^6 3s^2 3p^6$ . Since nuclear charge increases across the period, electrons are more strongly attracted to the nucleus. **[1]** Hence, cationic radius decreases from  $\text{Na}^+$  to  $\text{Si}^{4+}$ , and anionic radius decreases from  $\text{P}^{3-}$  to  $\text{Cl}^-$ .

Anions have one more quantum shell of electrons than the cations.

**[1]**. Hence, there is a sharp increase in ionic radius from  $\text{Si}^{4+}$  to  $\text{P}^{3-}$ .

**Melting point**

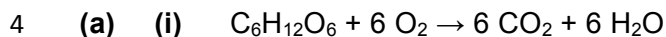
Na, Mg and Al have giant metallic structure. A lot of energy is needed to overcome the strong metallic bonds. Hence, they have high melting points.

Si has giant molecular structure. A lot of energy is needed to overcome the strong covalent bonds throughout the whole structure. Hence, melting point of Si is high.

P, S, Cl and Ar have simple molecular structure. A small amount of energy is needed to overcome the weak intermolecular van der Waals' forces of attraction. Hence, they have low melting points.

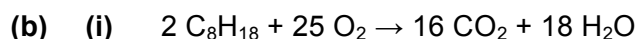
All 3 structures and corresponding bonds correct – **[2]**

At least 1 structure and corresponding bond correct – **[1]**



(ii)  $\text{no. of moles of glucose} = \frac{1.20 \times 1000}{180} = 6.67 \text{ mol}$  **[1]**  
 $\text{no. of moles of CO}_2 = 6 \times 6.67 = 40 \text{ mol}$  **[1]**

(iii)  $\text{total no. of moles of CO}_2$   
 $= 40 \times 6.82 \times 10^9 = 2.73 \times 10^{11} \text{ mol}$  **[1]**  
 $\text{mass of CO}_2$   
 $= 2.73 \times 10^{11} \times 44 = 1.20 \times 10^{13} \text{ g} = 1.20 \times 10^7 \text{ tonnes}$  **[1]**



(ii)  $\text{mass of octane} = 0.7 \times 4 \times 1000 = 2800 \text{ g}$  **[1]**  
 $\text{no. of moles of octane} = \frac{2800}{114} = 24.6 \text{ moles}$  **[1]**

(iii)  $\text{no. of moles of CO}_2 = 8 \times 24.6 = 197 \text{ moles}$  **[1]**  
 $\text{mass of CO}_2 = 197 \times 44 = 8660 \text{ g}$  **[1]**

(c)  $\text{No. of } 100 \text{ km} = \frac{1.20 \times 10^{13}}{8660} = 1.39 \times 10^9$  **[1]**

$\text{No. of km} = 1.39 \times 10^9 \times 100 = 1.39 \times 10^{11} \text{ km}$  **[1]**

[2]

[2]

[1]

[2]

[2]

[1]

[2]

[2]

[2]

[Total:12]

|           |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |
|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 5         | (a) (i) | <p>Reactivity : <math>\text{CH}_3\text{CH}_2\text{Cl} &lt; \text{CH}_3\text{CH}_2\text{Br} &lt; \text{CH}_3\text{CH}_2\text{I}</math> [1]<br/> As the size of the halogen atom increases (<math>\text{Cl} &lt; \text{Br} &lt; \text{I}</math>), the effectiveness of the overlapping of the p orbitals of the carbon and halogen atoms decreases. [1]<br/> This leads to a decrease in the strength of the C-X bond (<math>\text{C-Cl} &gt; \text{C-Br} &gt; \text{C-I}</math>). As the bonds get easier to break, the reactivity towards ammonia increases. [1]</p>                                                                                                                                                                                                                                                                           | [3]                          |
|           | (ii)    | Concentrated $\text{NH}_3$ , heat in a sealed tube                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | [1]                          |
|           | (iii)   | <div data-bbox="446 526 662 638" data-label="Chemical-Block"> <p style="text-align: center;">[1]</p> </div> <p>To minimise repulsion between the 3 bond pairs and 1 lone pair of electrons, the electron pairs arrange themselves tetrahedrally. [1]<br/> Shape: trigonal pyramidal; Bond angle: <math>107^\circ</math> [1]<br/> A weak Bronsted base is a <b>proton acceptor</b> [1] which undergoes <b>partial dissociation</b> [1] in water.<br/> (ii) <math>\text{CH}_3\text{CH}_2\text{NH}_3^+</math><br/> (iii) <math>K_b = 10^{-3.36} = 4.37 \times 10^{-4} \text{ mol dm}^{-3}</math> [1]<br/> <math>[\text{OH}^-] = \sqrt{4.37 \times 10^{-4} \times 0.01} = 2.09 \times 10^{-3} \text{ mol dm}^{-3}</math> [1]<br/> <math>\text{pOH} = -\lg 2.09 \times 10^{-3} = 2.68</math><br/> <math>\text{pH} = 14 - 2.68 = 11.3</math> [1]</p> | <br>[3]<br>[2]<br>[1]<br>[3] |
|           | (c) (i) | <p>In solution <b>B</b>, ethylamine is in excess while hydrochloric acid is in excess in solution <b>C</b>. [1]<br/> Solution <b>B</b> comprises of a weak base, ethylamine and its salt, <math>\text{CH}_3\text{CH}_2\text{NH}_3^+</math>. Hence, it is a buffer. [1]<br/> Solution <b>C</b> comprises of a strong acid, <math>\text{HCl}</math> and its salt, <math>\text{CH}_3\text{CH}_2\text{NH}_3^+\text{Cl}^-</math>. Hence, it is not a buffer. [1]</p>                                                                                                                                                                                                                                                                                                                                                                                | [3]                          |
|           | (ii)    | <p>When small amounts of acid is added,<br/> <math>\text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}^+ \rightarrow \text{CH}_3\text{CH}_2\text{NH}_3^+</math> [1]<br/> When small amounts of base is added,<br/> <math>\text{CH}_3\text{CH}_2\text{NH}_3^+ + \text{OH}^- \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}_2\text{O}</math> [1]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | [2]                          |
|           | (d)     | <p><math>2 \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-</math><br/> When temperature increases, the pH decreases which indicates that the concentration of <math>\text{H}^+</math> increases. This suggests that the equilibrium had shifted to the right, favouring the forward reaction. [1]<br/> Hence, the forward reaction is endothermic and an increase in temperature favours the ionisation of water. [1]</p>                                                                                                                                                                                                                                                                                                                                                                                             | [2]                          |
| Total: 20 |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |
|           |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |
| 6         | (a) (i) | <p>Order of reaction is defined as the power to which the concentration of that reactant is raised in an experimentally determined rate equation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | [1]                          |

(ii)

| Time / s | [RCOOH] / mol dm <sup>-3</sup> | [ester] / mol dm <sup>-3</sup> |
|----------|--------------------------------|--------------------------------|
| 0        | 0                              | 0.200                          |
| 25       | 0.048                          | 0.152                          |
| 50       | 0.085                          | 0.115                          |
| 75       | 0.112                          | 0.088                          |
| 100      | 0.133                          | 0.067                          |
| 125      | 0.149                          | 0.051                          |

Plot [ester] against time. [2]

Show at least 2 half-lives on graph. [1]

Constant half-life (approx. 63 s) [1]

[4]

(iii) The concentration of water is constant as water is the solvent and in large excess (pseudo-zero order w.r.t. water).

[1]

(iv) Let x be the order of reaction w.r.t. hydrochloric acid  
Comparing experiments 1 & 2,

$$\frac{rate_1}{rate_2} = \frac{k[0.10][0.10]^x}{k[0.15][0.20]^x} = \frac{1}{3}$$

$$x = 1$$

Order of reaction w.r.t. hydrochloric acid = 1

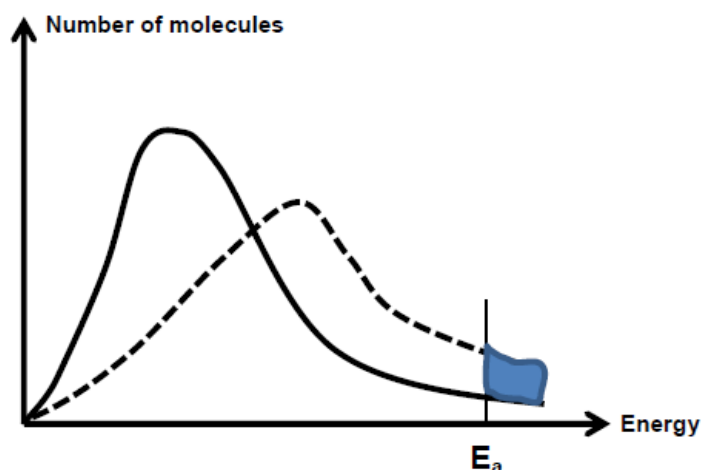
[1]

(v)  $Rate = k[ester][HCl]$  [1]

Units: mol<sup>-1</sup> dm<sup>3</sup> s<sup>-1</sup> [1]

[2]

(vi)



(Correct diagram – [1])

When temperature increases, the average energy of the reacting particles is increased. [1]

Hence, the proportion of particles with energy equal to or greater than the activation energy increases significantly as seen in the shaded area in the diagram. [1]

In addition, when the temperature of the reactants is increased, their average speeds increase and so, the frequency of collisions between reacting particles increases. This increases the number of effective collisions and consequently the rate of reaction. [1]

[4]

(b)  $K_c = \frac{[RCOOH][CH_3OH]}{[RCOOCH_3]} = \frac{0.125 \times 0.376}{0.188} = 0.25 \text{ mol dm}^{-3}$  [1]

|                       |                 |              |                      |              |                 |
|-----------------------|-----------------|--------------|----------------------|--------------|-----------------|
|                       | $RCOOCH_3 (aq)$ | $+ H_2O (l)$ | $\rightleftharpoons$ | $RCOOH (aq)$ | $+ CH_3OH (aq)$ |
| eqm conc              | $0.188 + x$     |              |                      | $0.200$      | $0.376 - x$     |
| /mol dm <sup>-3</sup> |                 |              |                      |              |                 |

$$K_c = \frac{0.200(0.376 - x)}{0.188 + x} = 0.25$$

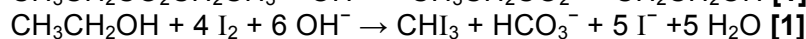
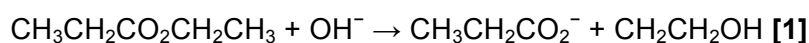
$$x = 0.0627$$
 [1]

Hence,  $[CH_3OH] = 0.376 - 0.0627 = 0.313 \text{ mol dm}^{-3}$  [1]

[3]

(c) Test: Warm with aq. I<sub>2</sub> and aq. NaOH. [1]

A yellow precipitate will be formed with ethyl propanoate while no precipitate will be formed with propyl ethanoate. [1]



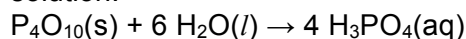
[4]

Total: 20

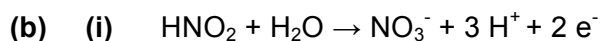
7 (a) MgO dissolves slightly in water to give a weakly alkaline solution.  
 $MgO(s) + H_2O(l) \rightleftharpoons Mg(OH)_2(aq)$

SiO<sub>2</sub> does not dissolve in water.

P<sub>4</sub>O<sub>10</sub> undergoes hydrolysis with water to give a strongly acidic solution.



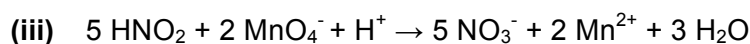
[4]



[1]

(ii) Oxidation state of N increase from +3 in HNO<sub>2</sub> to +5 in NO<sub>3</sub><sup>-</sup>.  
Hence, nitrogen undergoes oxidation and act as a reducing agent.

[2]



(iv)  $no. \text{ of moles of } KMnO_4 = 0.01 \times \frac{12.5}{1000} = 1.25 \times 10^{-4} \text{ mol}$

$$no. \text{ of moles of } NaNO_3 = no. \text{ of moles of } HNO_3$$

$$= \frac{5}{2} \times 1.25 \times 10^{-4} = 3.125 \times 10^{-4} \text{ mol}$$
 [1]

$$mass \text{ of } NaNO_3 = 3.125 \times 10^{-4} \times 85 = 0.0266 \text{ mol}$$

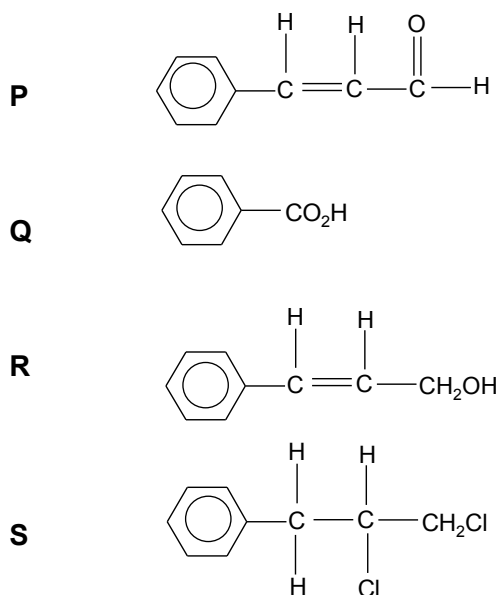
$$\% \text{ mass} = 0.0266 \times 100 = 2.66 \%$$
 [1]

[2]

(c)

| Information                                                                                                         | Deductions                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Orange ppt formed when <b>P</b> is formed with 2,4-DNPH                                                             | <b>P</b> is a carbonyl compound and it undergoes condensation.                                                                                                              |
| <b>P</b> gives a red ppt with Fehling's solution                                                                    | <b>P</b> is an aldehyde.                                                                                                                                                    |
| <b>P</b> decolourises aq. Br <sub>2</sub>                                                                           | <b>P</b> is an alkene and it undergoes addition with Br <sub>2</sub> .                                                                                                      |
| <b>Q</b> and CO <sub>2</sub> is formed when <b>P</b> is refluxed with MnO <sub>4</sub> <sup>-</sup> /H <sup>+</sup> | <b>P</b> undergoes side chain oxidation to form <b>Q</b> , benzoic acid.<br>OR <b>P</b> is an alkene and undergoes oxidative cleavage to form <b>Q</b> , a carboxylic acid. |
| <b>Q</b> is soluble in sodium hydroxide and sodium carbonate.                                                       | <b>Q</b> is a carboxylic acid and it undergoes neutralisation.                                                                                                              |
| Treatment of <b>P</b> with NaBH <sub>4</sub> gives <b>R</b> .                                                       | The aldehyde group in <b>P</b> is reduced to a primary alcohol in <b>R</b> .                                                                                                |
| <b>R</b> gives H <sub>2</sub> with Na.                                                                              | <b>R</b> is an alcohol and it undergoes redox reaction with Na.                                                                                                             |
| <b>R</b> reacts with HCl to form <b>S</b> .                                                                         | <b>R</b> contains alkene and alcohol groups and undergoes addition and substitution reactions respectively. <b>S</b> is a halogen derivative.                               |

Each deduction [1] Maximum 6M



Each structure [1]

[10]  
Total:20